

1.8.8 Tilings

W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W

W
B

W	B
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Figure 3: Two dominos

Figure 2: The standard checkerboard

A checkerboard is a rectangle divided into squares of the same size by horizontal and vertical lines. The game of checkers is played on a board with 8 rows and 8 columns; this board is called the standard checkerboard and is shown in Figure 2.

A domino is a rectangular piece that is one square by two squares as shown in Figure 3. We say that a board is tiled by dominos when all its squares are covered with no overlapping dominos and no dominos overhanging the board.

Example 18: Can we tile the standard checkerboard using dominos?

Solution: Yes, we can. Here is an example.

W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W
W	B	W	B	W	B	W	B
B	W	B	W	B	W	B	W

We tile each row with 4 horizontal dominos.
 $\therefore$ There will be $8 \times 4 = 32$ dominos needed.